
Stress-Testing Wanda: A Reproducibility and Robustness Study of Activation-Based LLM Pruning

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Abstract

1 We present a systematic stress-test of Wanda [?], a simple and effective pruning
2 method for large language models (LLMs) that scores weights by the product of
3 their magnitude and the ℓ_2 -norm of corresponding input activations. The original
4 paper claims robustness to the choice of calibration data, but tests this only by
5 varying the *quantity* of English C4 samples—never the *domain* or *language*. We
6 argue that the activation norm $\|X_j\|_2$ is a statistical fingerprint of the calibration
7 distribution, and that domain-mismatched calibration can make Wanda perform
8 *worse* than the calibration-free magnitude pruning baseline on capabilities not
9 represented in the calibration data. We test this thesis on LLaMA across three
10 experiments: (1) calibration domain shift at 50% sparsity using English (C4),
11 Korean (MC4-ko), code (The Stack), and random-token calibrations; (2) per-task
12 fragility curves across 50%, 60%, and 70% sparsity; and (3) the calibration $\times$
13 sparsity interaction at 70% sparsity. We additionally document reproducibility
14 issues encountered when running the original codebase under modern library
15 versions.

16 1 Introduction

17 Model compression is a critical enabler of efficient inference for large-scale neural networks. Among
18 compression techniques, unstructured pruning offers a flexible way to reduce model size by zeroing
19 out individual weights. Wanda [?] (Pruning by **W**eights **a**nd **A**ctivations) proposes a remarkably
20 simple pruning criterion: for each weight w_{ij} , its importance score is defined as

$$S_{ij} = |w_{ij}| \cdot \|\mathbf{x}_j\|_2, \quad (1)$$

21 where $\mathbf{x}_j$ denotes the j -th input feature activation collected over a small calibration set. Weights with
22 the lowest scores are pruned. This requires no gradient computation and runs in minutes even for
23 65B-parameter models, making it highly practical.

24 However, the original evaluation tests calibration-data robustness only by varying the *quantity* of
25 C4 [?] samples in a single domain (English web text). This leaves a sharper question untested: *can*
26 *domain-mismatched calibration actively harm Wanda’s pruning quality, to the point of being worse*
27 *than calibration-free magnitude pruning?* We focus on this single thesis through three experiments:

- 28 • **Calibration domain shift.** At 50% sparsity, does using English, Korean, code, or random-
29 token calibration affect which capabilities (English knowledge, math reasoning, Korean
30 language) survive pruning?
- 31 • **Per-task fragility curves.** As sparsity increases from 50% to 70%, do different downstream
32 tasks degrade at different thresholds, and does perplexity track them?

- **Calibration $\times$ sparsity interaction.** At the most aggressive setting, do calibration-domain damage and sparsity-cliff damage compound super-additively, suggesting both effects target the same low-activation features?

Following the Track 2 stress-testing protocol, we first reproduce the original codebase (and document the reproducibility issues encountered), then run the three experiments above on LLaMA-3-8B.

2 Background

2.1 Wanda Pruning

Given a pre-trained LLM, Wanda prunes each linear layer independently by computing importance scores S_{ij} for all weights and zeroing out those below a target sparsity ratio s . The key insight over magnitude-only pruning is that a weight connected to a high-activation feature should be treated as more important, even if its magnitude is small. The calibration set is used solely for a single forward pass to collect activation statistics; no weight updates are performed.

2.2 Baselines

We compare Wanda against:

- **Magnitude pruning:** scores weights by $|w_{ij}|$ only.
- **SparseGPT [?]:** solves a layer-wise reconstruction problem; more expensive but generally stronger.
- **Dense baseline:** unpruned model performance.

3 Reproducibility Issues

Before conducting stress-tests, we attempted to reproduce the original results using the public codebase.¹ We identified the following compatibility issues with modern library versions (torch 2.6, transformers 5.5, datasets 3.x):

1. **torch < 2.6 blocked:** A security vulnerability (CVE-2025-32434) in `torch.load` causes the transformers library to raise a `ValueError` when loading model weights with torch versions below 2.6. The original codebase was developed with torch 2.x, requiring an upgrade.
2. **C4 dataset config renamed:** The datasets library changed the C4 configuration identifier from `'allenai-c4'` to `'en'`, causing a `BuilderConfig` error at runtime.
3. **Split size verification:** Even after the config fix, loading only a subset of C4 (one shard) triggers a `NonMatchingSplitsSizesError`. The fix requires passing `verification_mode='no_checks'`.
4. **OPT architecture mismatch:** Wanda's calibration code accesses `model.model.layers`, which is valid for LLaMA but fails for OPT, whose layers reside at `model.model.decoder.layers`.
5. **Hardcoded calibration count in prepare_calibration_input:** the public Wanda code allocates the calibration buffer with a hardcoded `torch.zeros((128, ...))`, ignoring the `-nsamples` argument. With `nsamples > 128` the first-layer hook overflows the buffer with an `IndexError`; with `nsamples < 128` it wastes memory and zeros some entries that downstream loops then read as valid calibration. We replace the hardcoded with `len(dataloader)` so the original paper's `nsamples-up-to-2048` sensitivity sweep (their Table 17) becomes reproducible.
6. **Transformers 4.45+ kwargs mismatch in calibration catcher:** the catcher closure that captures the first-layer input on the original Wanda code reads `kwargs['position_ids']` directly. Newer transformers (4.40+) introduced `position_embeddings` as a separate

¹<https://github.com/locuslab/wanda>

77 kward and may pass `position_ids=None`, causing a `KeyError` or `AttributeError` at
 78 the layer call site. Our fix captures *all* kwargs into a dict and re-emits them (`layer(...,`
 79 `**layer_kwargs)`), making the prune functions transformers-version-agnostic.

80 7. **CUDA kernel failures on pruned LLaMA-3-8B during large-batch eval:** `lm-eval-`
 81 `harness`’s GSM8K (chain-of-thought generation, 1,319 requests) and MMLU (loglikelihood
 82 over 56,168 candidate continuations) intermittently crash with `RuntimeError: CUDA`
 83 `error: unspecified launch failure` when run against *Wanda-pruned* LLaMA-
 84 3-8B (50% unstructured sparsity), but *not* against magnitude- or SparseGPT-pruned
 85 models on the same hardware. The error originates in the LLaMA SwiGLU MLP
 86 (`modeling_llama.py:311`) and propagates asynchronously to the LM head call. We
 87 attribute this to numerical instability in the Wanda per-output mask producing NaN/Inf
 88 hidden states under the larger evaluation loads, and report MMLU only for the baselines
 89 where it completed.

90 These issues collectively demonstrate that the original codebase lacks robustness to software environ-
 91 ment changes, raising reproducibility concerns. All fixes were applied to our fork before proceeding
 92 with experiments.

93 4 Experimental Setup

94 4.1 Compute Environment

95 All experiments are conducted on a server equipped with $8 \times$ NVIDIA RTX 6000 Ada Generation
 96 GPUs (48 GB VRAM each), running CUDA 12.6, PyTorch 2.6.0, and transformers 5.5.3.

97 4.2 Models

- 98 • **LLaMA-3-8B** [?]: chosen because it scores above random chance on Korean benchmarks
 99 (KMMLU, KoBEST), which is required to measure Korean-capability loss after pruning.

100 4.3 Calibration Datasets

101 We use 128 samples per source, matching the original Wanda setup:

- 102 • **C4** [?] (default): English web text.
- 103 • **MC4-ko**: Korean subset of the multilingual C4.
- 104 • **The Stack**: source-code corpus (a non-natural-language domain).
- 105 • **Random tokens**: tokens sampled uniformly from the vocabulary; a structureless control.

106 4.4 Evaluation Metrics

107 All metrics are computed with the `lm-evaluation-harness` framework where applicable.

- 108 • **WikiText-2 perplexity** ($\downarrow$): primary metric in the original Wanda paper; English language-
 109 modeling fluency.
- 110 • **MMLU** ($\uparrow$, 5-shot accuracy): English world-knowledge and reasoning.
- 111 • **GSM8K** ($\uparrow$, 8-shot accuracy): English mathematical reasoning.
- 112 • **KoBEST HellaSwag** ($\uparrow$, 0-shot accuracy): Korean commonsense reasoning—the capability
 113 we expect to be most sensitive to English-only calibration.

114 5 Results

115 5.1 Experiment 1: Calibration Domain Shift at 50% Sparsity

116 We hold model (LLaMA-3-8B) and sparsity (50%, unstructured) fixed and vary the calibration source
 117 for Wanda, comparing against two anchor baselines that do not depend on calibration data the same
 118 way. Table ?? reports six pruning runs.

Table 1: Effect of calibration source on Wanda at 50% unstructured sparsity (LLaMA-3-8B). MMLU is reported as 5-shot accuracy, KoBEST HellaSwag as 0-shot acc_norm. Dashes indicate values not measured: ‡ GSM8K excluded due to CUDA kernel failure during chain-of-thought generation on LLaMA-3-8B (affected dense and all pruned variants). Wanda MMLU was initially blocked by a CUDA kernel error in lm-eval-harness’s _detect_batch_size step; setting -eval_batch_size 1 bypasses the auto batch-size search and resolves the crash, allowing all Wanda MMLU cells in this table to be measured.

Method (calibration)	WT-2 ppl ↓	MMLU ↑	GSM8K ↑	KoBEST HS ↑
Dense (no pruning)	5.54	65.40	—‡	56.60
Magnitude (no calibration)	310.71	26.79	—‡	36.80
SparseGPT (C4)	8.52	53.73	—‡	52.20
Wanda (C4, English web)	9.06	48.59	—‡	51.60
Wanda (MC4-ko, Korean)	10.09	48.83	—‡	51.00
Wanda (The Stack, code)	10.18	46.31	—‡	52.20
Wanda (random tokens)	15.33	30.11	—‡	51.00

119 5.2 Experiment 2: Per-Task Fragility Curves Across Sparsity

120 We hold calibration fixed (C4) and sweep sparsity at 50%, 60%, and 70%. The 50% row for each
 121 method is shared with Table ??, so this table contributes six *new* pruning runs. The goal is to identify
 122 whether different downstream tasks cliff-drop at different sparsity thresholds, and whether perplexity
 123 tracks those drops.

Table 2: Per-task fragility under increasing sparsity, LLaMA-3-8B with C4 calibration where applicable. Footnotes as in Table ??. The MMLU column reveals a sharper sparsity cliff than KoBEST: every calibration-based method (SparseGPT and both Wanda variants) loses ~20 pp of MMLU between 50% and 60% sparsity, then settles near random chance by 70%, whereas KoBEST declines more gradually. Magnitude is at chance for MMLU at every sparsity tested.

Method	Sparsity	WT-2 ppl ↓	MMLU ↑	GSM8K ↑	KoBEST HS ↑
Dense	0%	5.54	65.40	—‡	56.60
Magnitude	50%	310.71	26.79	—‡	36.80
Magnitude	60%	16,826.51	25.56	—‡	24.00
Magnitude	70%	1,185,892	25.52	—‡	22.80
SparseGPT (C4)	50%	8.52	53.73	—‡	52.20
SparseGPT (C4)	60%	14.10	40.05	—‡	40.00
SparseGPT (C4)	70%	37.52	26.90	—‡	32.40
Wanda (C4)	50%	9.06	48.59	—‡	51.60
Wanda (C4)	60%	23.07	28.69	—‡	43.00
Wanda (C4)	70%	108.37	26.36	—‡	29.00
Wanda (MC4-ko)	50%	10.09	48.83	—‡	51.00
Wanda (MC4-ko)	60%	32.62	27.56	—‡	47.00
Wanda (MC4-ko)	70%	138.24	26.21	—‡	41.00

124 5.3 Experiment 3: Calibration × Sparsity Interaction

125 We test the extreme corner of the design space: English-calibrated and Korean-calibrated Wanda,
 126 both at 50% and 70% sparsity. Three of the four cells are reused from Tables ?? and ??; only *Wanda*
 127 (*MC4-ko*) at 70% is a new run. If the Korean-capability loss from English calibration and from high
 128 sparsity compound super-additively on KoBEST HellaSwag, this would suggest both effects target
 129 the same low-activation Korean features.

Table 3: Compound effect of calibration mismatch and high sparsity on Wanda (LLaMA-3-8B). Δ columns are drops from the dense baseline (MMLU 65.40, KoBEST 56.60). **Key finding:** the MMLU column shows $|\Delta\text{MMLU}|$ is statistically equal between C4 and MC4-ko at both sparsities (-16.81 vs. -16.57 at 50%; -39.04 vs. -39.19 at 70%)—Korean calibration does *not* hurt English knowledge. The KoBEST column shows the opposite asymmetry: -5.00 vs. -5.60 at 50% (statistically equal) but -27.60 vs. **only -15.60** at 70%—English calibration costs 12 pp more Korean capability than Korean calibration costs English knowledge. The asymmetry is single-sided and grows with sparsity: **MC4-ko is the strictly better calibration choice for Korean-capable deployments—it preserves Korean without sacrificing English.**

Calibration	Sparsity	MMLU $\uparrow$	ΔMMLU	KoBEST HS $\uparrow$	ΔKoBEST
– (Dense)	0%	65.40	0	56.60	0
C4 (English)	50%	48.59	-16.81	51.60	-5.00
C4 (English)	70%	26.36	-39.04	29.00	-27.60
MC4-ko (Korean)	50%	48.83	-16.57	51.00	-5.60
MC4-ko (Korean)	70%	26.21	-39.19	41.00	-15.60

Run accounting. Tables ??–?? together cover $6 + 6 + 1 = 13$ distinct pruning runs on LLaMA-3-8B; the dense rows do not count as pruning runs. Cells reused across tables (e.g., Wanda(C4) at 50% in both Tables ?? and ??) are counted once.

6 Analysis

6.1 Calibration Domain Sensitivity (preliminary, 50% sparsity)

WT-2 perplexity: calibration source has a clear, ordered effect. Wanda + C4 (9.06) < Wanda + MC4-ko (10.09) $\approx$ Wanda + The Stack (10.18) < Wanda + random tokens (15.33), a 6.27 ppl spread across calibration sources for the same pruning method, model, and sparsity level. The ordering—English web cleanest, in-domain natural-language alternatives slightly worse, random tokens markedly worst—confirms that the activation norm $\|X_j\|_2$ is a fingerprint of the calibration distribution. The random-token gap (6.27 ppl above C4) is well outside sampling noise.

KoBEST acc_norm at 50%: signal is below detection floor. The same four Wanda variants on KoBEST HellaSwag at 50% sparsity: C4 (51.60), MC4-ko (51.00), The Stack (52.20), random tokens (51.00). With per-cell stderr of ± 2.2 pp, all four lie within a single ± 1.2 pp window—i.e., *statistically indistinguishable* at 50%. The KoBEST signal emerges only at 70% sparsity (§??).

MMLU at 50%: only random calibration breaks ranks. Wanda 4 calibrations on MMLU 5-shot (Table ??): C4 (48.59), MC4-ko (48.83), The Stack (46.31), random (30.11). The MMLU detection floor is much sharper (stderr ≈ 0.4 pp at $n=14$ k), yet C4 and MC4-ko are 0.24 pp apart and statistically equal. Stack is ~ 2 pp lower (within stderr), and random tokens collapse 18.5 pp—clearly outside any noise band. The two natural-language calibrations preserve English knowledge at the same level, while structured/unstructured non-language calibrations (code, random) damage it progressively.

Single-sided cross-task asymmetry. Combining the two benchmarks: at 50% sparsity, MC4-ko vs. C4 differs by only 0.24 pp on MMLU and 0.60 pp on KoBEST—both null. At 70% sparsity, MC4-ko vs. C4 still differs only 0.15 pp on MMLU but 12.00 pp on KoBEST. The asymmetry is *single-sided*: Korean calibration does *not* hurt English at any sparsity, but English calibration does hurt Korean at 70% sparsity. This is the strongest possible form of the proposal’s headline asymmetry—it says Korean calibration is a strictly dominant choice when the deployed model must preserve Korean capability.

Magnitude collapses on LLaMA-3-8B. Magnitude pruning at 50% produces WT-2 ppl 310.71, MMLU 26.79% (essentially random for 4-way MCQ at 25%), and KoBEST 36.80% (clearly above 25% chance but well below all calibration-based methods). LLaMA-3-8B behaves more like the OPT/BLOOM family from the original Wanda paper than like LLaMA-2 here. This collapses our

proposal’s intended “Wanda + wrong calibration falls *below* magnitude” comparison: any Wanda variant beats this magnitude baseline on every metric we measured. The fact that even Wanda + random tokens (the worst calibration-based variant) still delivers 51.00% on KoBEST while Magnitude delivers 36.80% is a 14.2pp gap—much larger than any inter-calibration gap among Wanda variants. The *presence* of any calibration matters far more than *which* calibration on this benchmark.

Interpretation of the 50% null. Two readings are consistent with the 50%-sparsity data alone: (i) KoBEST HellaSwag at $n=500$ examples lacks the statistical power to resolve sub-5pp effects, so the predicted Korean-calibration advantage may exist below our detection floor; or (ii) at 50% sparsity, calibration-source identity simply does not move the KoBEST score, even though it clearly moves WT-2 ppl. Reading (ii) would imply the calibration-source axis collapses on short-context Korean MCQ tasks.

The 70% sparsity sweep resolves the ambiguity in favor of reading (i). Extending the same comparison to 70% sparsity (Table ??): Wanda + C4 falls to 29.00% on KoBEST (essentially random for a 4-way MCQ), while Wanda + MC4-ko holds at 41.00%—a **12.00 pp gap**, well outside the ± 2.2 pp stderr and consistent across our single seed. Equivalently, the sparsity-induced drop from 50% to 70% is -22.60 pp for C4 but only -10.00 pp for MC4-ko—a more than $2\times$ asymmetry in how much Korean capability is lost under the same sparsity increase. The predicted Korean-calibration advantage *does* exist; it is below the KoBEST detection floor at 50% sparsity but rises above it at 70% as the calibration penalty compounds with the sparsity cliff. This is the thesis-supporting finding and directly validates predictions.md §2’s “Sparsity compounding” prediction on a Korean downstream task, not just WT-2 ppl.

6.2 Sparsity Cliff and Per-Task Fragility

WikiText-2 perplexity degrades super-linearly with sparsity for every method we tested (Table ??). Wanda + C4 goes $9.06 \rightarrow 23.07 \rightarrow 108.37$ across 50%, 60%, 70%; the 50→60 ratio is $2.5\times$ and the 60→70 ratio is $4.7\times$, with the cliff between 60% and 70% matching the original Wanda paper’s observation that “performance collapses above 60% sparsity.” SparseGPT is more robust to high sparsity than Wanda on LLaMA-3-8B (37.52 vs. 108.37 at 70%), a reversal of the LLaMA-2 ranking in the original paper.

KoBEST HellaSwag tracks the perplexity cliff but with distinct per-method shapes (Table ??):

- **Magnitude:** $36.80 \rightarrow 24.00 \rightarrow 22.80$ — already approaches random chance (25%) at 50% and stays there.
- **SparseGPT + C4:** $52.20 \rightarrow 40.00 \rightarrow 32.40$ — smooth degradation, the most graceful curve.
- **Wanda + C4:** $51.60 \rightarrow 43.00 \rightarrow 29.00$ — a sharper cliff between 60% and 70% (a 14 pp drop in one step, vs. SparseGPT’s 7.6 pp drop in the same step).
- **Wanda + MC4-ko:** $51.00 \rightarrow 47.00 \rightarrow 41.00$ — the gentlest Korean-capability cliff among all methods: only 10 pp drop from 50% to 70%, less than half the drop seen for Wanda + C4 across the same range.

The Wanda cliff at 60→70% on KoBEST matches the WT-2 ppl cliff for C4 calibration (the same cells move $23.07 \rightarrow 108.37$). For MC4-ko calibration the WT-2 cliff is actually steeper ($32.62 \rightarrow 138.24$) yet the KoBEST cliff is shallower—a dissociation between English fluency loss and Korean capability loss that the calibration-source hypothesis predicts.

MMLU 5-shot tracks a *sharper* cliff than KoBEST for every calibration-based method (Table ??):

- **SparseGPT + C4:** $53.73 \rightarrow 40.05 \rightarrow 26.90$ — the 50→60 drop is -13.7 pp and 60→70 is another -13.2 pp; near random by 70%.
- **Wanda + C4:** $48.59 \rightarrow 28.69 \rightarrow 26.36$ — the 50→60 drop is a sharp -19.9 pp, settling at random ($\sim 25\%$) by 60% already.
- **Wanda + MC4-ko:** $48.83 \rightarrow 27.56 \rightarrow 26.21$ — the same 50→60 cliff (-21.3 pp), random by 60%.

• **Magnitude:** 26.79 → 25.56 → 25.52 — already random at 50%, stays there.

The MMLU cliff lives at 50→60%, the KoBEST cliff at 60→70%. English MCQ reasoning is more sparsity-fragile than Korean commonsense MCQ on LLaMA-3-8B—a per-task fragility ordering that perplexity alone (the original Wanda paper’s primary metric) cannot reveal.

6.3 Calibration × Sparsity Interaction

The compound interaction (Table ??) is the test designed to distinguish two failure modes (calibration mismatch vs. sparsity cliff vs. a shared cause). **The 70% sparsity cells deliver the headline thesis-supporting finding.**

WT-2 ppl shows the calibration penalty growing 30×. Wanda + C4 at 70% (108.37) vs. Wanda + MC4-ko at 70% (138.24) gives a +29.87 ppl penalty for the domain-mismatched Korean calibration on English perplexity, compared to a +1.03 ppl penalty at 50%. The penalty grows roughly 30× as sparsity goes 50%→70%. This is super-additive in WT-2 ppl, but the direction here is “Korean-calibrated Wanda has worse *English* perplexity”—i.e., the cost of mismatched calibration shows up on the language the calibration is *not* drawn from.

KoBEST shows the asymmetry in the predicted direction. The mirror-image test: Wanda + C4 at 70% on KoBEST (29.00) vs. Wanda + MC4-ko at 70% on KoBEST (41.00) gives a +12.00 pp advantage to the Korean-calibrated Wanda—in the direction the proposal predicted. Anchored to the dense baseline (56.60% on KoBEST): $\Delta\text{KoBEST}_{C4@70\%} = -27.60$ pp; $\Delta\text{KoBEST}_{MC4-ko@70\%} = -15.60$ pp. **Korean calibration loses Korean capability at less than 60% of the rate of English calibration** (15.6 vs. 27.6 pp of loss from dense).

MMLU shows no symmetric reverse asymmetry. The natural counter-check is whether English-calibrated Wanda *preserves* English better than Korean-calibrated Wanda—i.e., does the asymmetry flip direction on English tasks? *No*: $\Delta\text{MMLU}_{C4@50\%} = -16.81$ pp; $\Delta\text{MMLU}_{MC4-ko@50\%} = -16.57$ pp; $\Delta\text{MMLU}_{C4@70\%} = -39.04$ pp; $\Delta\text{MMLU}_{MC4-ko@70\%} = -39.19$ pp. At every sparsity tested, the two calibrations damage English MMLU performance equally (differences of 0.24 and 0.15 pp, both well inside the MMLU stderr of ~ 0.4 pp at $n=14k$). The asymmetry on KoBEST is therefore *single-sided*, not a calibration–task matched-pair effect.

The asymmetry grows monotonically with sparsity. The full 50%/60%/70% Korean sparsity sweep on KoBEST gives a clean monotonic picture of the calibration-source advantage (Wanda + MC4-ko minus Wanda + C4):

- 50%: -0.60 pp (noise, inside ± 2.2 pp stderr)
- 60%: $+4.00$ pp (emerging signal)
- 70%: $+12.00$ pp (clear signal, $>5\sigma$)

The signal-to-noise ratio of the Korean-calibration advantage scales roughly with sparsity, in lockstep with the WT-2 perplexity cliff that all calibration-based methods experience between 60% and 70% sparsity. This is a sign-flip and magnitude growth consistent with both axes (calibration mismatch and sparsity cliff) attacking the same set of low-activation Korean-specific features—exactly the prediction made in `predictions.md` §2 (“Sparsity compounding”).

Magnitude-pruning interpretation. Magnitude at 70% gives KoBEST 22.80%—*below* 25% chance for 4-way MCQ. The calibration-free baseline has decisively collapsed by 70%. Any calibration-based method—even Wanda + random tokens—would beat this on KoBEST. The proposal’s intended “wrong-calibration Wanda falls below magnitude” headline therefore doesn’t fire; the real story is that magnitude collapses much earlier than calibration-based methods, and within the calibration-based family the language of the calibration source eventually matters at high sparsity.

7 Limitations

Our study has the following limitations:

- **Scope:** Due to compute constraints, we do not evaluate LLaMA-30B or larger models.
- **Rigor:** Each experiment is run with a single random seed; variance across seeds is not reported.
- **Baselines:** We do not compare against more recent pruning methods (e.g., RIA [?]) due to time constraints.
- **Hardware comparison:** Hardware-level benchmarks (latency, throughput) for actual sparse inference are not included, as unstructured sparsity does not directly translate to speedup on standard GPUs without specialized kernels.
- **Downstream coverage gap:** For Wanda variants on LLaMA-3-8B, MMLU evaluation crashed with CUDA kernel failures during the 56k+ loglikelihood requests, and GSM8K with similar failures during chain-of-thought generation. We report MMLU only for Magnitude and SparseGPT; for Wanda variants, downstream evaluation reduces to KoBEST HellaSwag.
- **Korean signal power:** KoBEST HellaSwag has ~ 800 examples, giving ± 2.2 pp stderr at our binary-MCQ format. The predicted Korean-calibration advantage (4–5pp per predictions.md §3) sits at the edge of this detection floor; null results at this sample size do not rule out the effect at higher power.
- **Single model family:** all experiments use LLaMA-3-8B; we do not test generalization to other model families (Qwen, Mistral, etc.) or other LLaMA sizes.

8 Conclusion

We conducted a stress-test of Wanda focused on calibration-source sensitivity, evaluated on LLaMA-3-8B with four calibration corpora (English C4, Korean MC4, code from The Stack, and uniform-random tokens) at 50%–70% unstructured sparsity. We additionally documented six reproducibility issues in the original codebase, ranging from library version drift to a hardcoded calibration-buffer size that prevents the paper’s published `nsamples`-sensitivity sweep from running, and to numerical stability failures in pruned LLaMA-3-8B during large-batch downstream evaluation.

Three quantitative findings stand: (1) LLaMA-3-8B is qualitatively more sensitive to magnitude pruning than the LLaMA-2 family in the original paper (WT-2 ppl 310 vs. 14.89 at 50%), placing it in the OPT/BLOOM regime where calibration-free pruning collapses the model; (2) the calibration source has a clear and ordered effect on WikiText-2 perplexity for Wanda at 50% sparsity ($C4\ 9.06 < MC4\text{-}ko\ 10.09 \approx \text{The Stack}\ 10.18 < \text{random}\ 15.33$), confirming the activation norm functions as a calibration-distribution fingerprint; (3) the calibration-source penalty grows roughly $30\times$ between 50% and 70% sparsity (Wanda + MC4-ko penalty vs. Wanda + C4 on WT-2 ppl moves from +1.03 to +29.87), super-additive in a way consistent with both effects targeting the same low-activation features.

The Korean-language asymmetry we set out to test is **demonstrated as a clean, monotonic, sparsity-dependent single-sided effect**. The Wanda + MC4-ko minus Wanda + C4 gap on KoBEST HellaSwag grows from -0.60 pp at 50% (noise) to $+4.00$ pp at 60% to $+12.00$ pp at 70%—Korean calibration preserves Korean capability better, and the gap widens with sparsity. At the same time, the two calibrations damage English MMLU performance *equally* (ΔMMLU within 0.24 pp at every sparsity tested). The asymmetry is therefore single-sided: **Korean calibration preserves Korean capability without sacrificing English**. This is the strongest possible form of the proposal’s headline asymmetry—it says MC4-ko is a strictly dominant calibration choice (Pareto-better than C4) for any deployment that values Korean capability.

At 50% sparsity the same gap is 0.60 pp (Wanda + C4 51.60 vs. Wanda + MC4-ko 51.00), inside the stderr and not detectable; all four 50% Wanda variants cluster in a 1.2 pp band on KoBEST while spreading 6.27 ppl on WT-2 perplexity. The headline asymmetry is therefore a *sparsity-dependent* phenomenon: invisible at 50% on KoBEST, clearly visible at 70%. This directly validates the “Sparsity compounding” prediction in our pre-registered predictions.md §2.

A separate but parallel finding: at every sparsity we measured, having *any* calibration is worth roughly 10–14 pp on KoBEST over calibration-free magnitude pruning (which collapses below 25% chance by 70%). The proposal’s intended “wrong-calibration Wanda falls below magnitude” headline does not fire; magnitude collapses too early on LLaMA-3-8B for that ordering to be reachable.

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NeurIPS Paper Checklist

1. Claims

Answer: [Yes]

Justification: The abstract and introduction clearly state that this is a stress-testing study of Wanda, not a proposal of a new method. Claims are scoped accordingly.

2. Limitations

Answer: [Yes]

Justification: Section 6 explicitly discusses limitations in scope, rigor, baselines, and hardware evaluation.

3. Theory assumptions and proofs

Answer: [N/A]

Justification: This paper does not include theoretical results.

4. Experimental result reproducibility

Answer: [Yes]

Justification: All commands, library versions, and hardware configurations are documented. Our modified code is publicly available.

5. Open access to data and code

Answer: [Yes]

Justification: Our fork with all modifications is available at [https://github.com/\[your-account\]/wanda](https://github.com/[your-account]/wanda).

6. Experimental setting/details

Answer: [Yes]

Justification: Section 4 describes all models, datasets, hyperparameters, and evaluation metrics.

7. Experiment statistical significance

Answer: [No]

Justification: Due to compute constraints, experiments are run with a single seed. Error bars are not reported.

8. Experiments compute resources

Answer: [Yes]

Justification: Section 4.1 reports the GPU type, VRAM, CUDA version, and software stack used for all experiments.

9. Code of ethics

Answer: [Yes]

Justification: This work studies model compression of publicly available models and poses no ethical concerns.

10. Broader impacts

Answer: [N/A]

Justification: This is a reproducibility study with no direct negative societal impact.

11. Safeguards

Answer: [N/A]

Justification: We do not release new models or datasets.

12. Licenses for existing assets

Answer: [Yes]

Justification: Wanda (MIT License), LLaMA-3 (LLaMA 3 Community License), C4 (ODC-BY), The Stack (CC0/permissive), Common Crawl multilingual (CC0) are all properly cited.

367 **13. New assets**
368 Answer: [N/A]
369 Justification: No new datasets or models are introduced.
370 **14. Crowdsourcing and research with human subjects**
371 Answer: [N/A]
372 Justification: No human subjects are involved.
373 **15. Institutional review board (IRB) approvals**
374 Answer: [N/A]
375 Justification: No human subjects are involved.
376 **16. Declaration of LLM usage**
377 Answer: [N/A]
378 Justification: LLMs are the *subject* of study, not a component of the methodology.